Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/03/2020
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Circulating miRNAs Provides Systemic Host Response to Microgravity: Utilizing GeneLab Datasets to Identify Molecular Targets for Spaceflight Studies		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	02/28/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2022 per F. Hernandez/ARC (Ed., 5/5/21) NOTE: End date changed to 9/30/2021 per F. Hernandez/ARC (Ed., 10/20/20)		
Key Personnel Changes/Previous PI:	Update on 12/2/2020: Dr. Egle Cekanaviciute will no longer be getting funds since she has become a civil servant. That money will go back to Dr. Beheshti for research to use on the grant. Ed. note 11/6/2019: CoInvestigators have changed per approval of NASA Ames--Dr. Peter Grabham replaces original CoI Dr. John McDonald and Dr. Egle Cekanaviciute is added as CoI to assist Dr. Charles Vanderburg who was a previous Co-I and now is a consultant on the grant.		
COI Name (Institution):	Blaber, Elizabeth Ph.D. (Rensselaer Polytechnic Institute grant 80NSSC20K0307 (formerly NASA ARC employee)) Grabham, Peter Ph.D. (Columbia University grant 80NSSC19K1658)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biological risks associated with microgravity are a major concern for long-term space travel. Although determination of risk has been a focus for NASA biological research, data examining systemic (i.e., multi- or pan-tissue) responses to spaceflight are sparse. For this proposal, we utilized the NASA GeneLab database to analyze the transcriptomic data from 7 different rodent tissue datasets. Using our systems biology expertise, we were able to reveal for the first time the existence of potential "master regulators" coordinating systemic responses to microgravity. We identified p53, Transforming Growth Factor Beta 1 (TGFB1), and immune related pathways as the highly prevalent pan-tissue signaling pathways that are affected by microgravity responses. Key genes/drivers isolated by our unbiased systems biology analyses displayed a major overlap between tissue types and established that TGFB1 was the most inter-connected gene across all datasets. Finally, a hypothesis was generated indicating that an overall systematic microgravity response driven by TGFB1 is caused by a circulating microRNA (miRNA) signature. This miRNA signature, consisting of 13 significantly expressed miRNAs, was predicted from differential gene expression that causes miRNA-driven microgravity-related functional states. A subsequent impact on health using a theoretical "health risk score" was calculated. The genes and miRNAs identified from our analyses will be used for the ground based studies which will be proposed for this solicitation and will be investigated to develop potential countermeasures against microgravity response. This proposal exemplifies the utility of the GeneLab data repository to aid in the process of performing novel hypothesis-based spaceflight research aimed at elucidating the global impact of environmental stressors at multiple biological scales.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This miRNA signature can be a minimally invasive biomarker to monitor health risk during deep space missions. The antagomir countermeasures being developed can act as a preventive measure against space radiation and microgravity effects. Countermeasures can be applied throughout mission to potentially mitigate any space environment effects. The same techniques can be used in the clinic to monitor various diseases such as muscle degeneration risk in patients. miRNA signature can be used during yearly checkup as another diagnostic blood based test to monitor risk for these diseases. If miRNA signature starts to appear in patients, then can apply antagomir treatment to suppress active miRNAs that are associated with disease progression long before the manifestation of the disease will occur.
Task Progress:	We have fully characterized the miRNA spaceflight signature and its role with different models that were exposed to the space environment. For experimental validation of the role of spaceflight-associated miRNAs in rodent models, we have investigated their changes in response to individual spaceflight stressors: microgravity using hindlimb unloading (HU), as well as ionizing radiation (IR) to simulate galactic cosmic rays using their chief components: protons and/or high mass and charge 56Fe particles. We have exposed male 16-week old mice to HU, IR, or their combination and measured miRNAs in their serum at 1 and 11 days post exposure using droplet digital PCR (ddPCR). These studies were finished and finalized in the beginning of 2020. Most miRNAs (with the exception of miR-217) were observed to increase with radiation dose and linear energy transfer (LET; gamma <150 MeV proton <600 MeV/n 56Fe) in mouse serum both at 1 day and 11 days post-IR. At 11 days post-IR, the only significant miRNA increases were observed in IR-only (no HU) mice irradiated with either 1Gy or 2Gy 600MeV/n 56Fe, or 1Gy protons . In combination, these results indicate that the miRNA signature is mainly driven by the particle radiation component, which is proportional to dose and LET. To examine the contribution of reduced gravity to the miRNA signature at higher resolution, a partial weight bearing (PWB) Wistar male rat model was used to simulate varying degrees of gravitational loading. We observed that decreasing amounts of gravitational loading increased the amount of miRNAs of interest, suggesting a gravity-dependent dose response. These results also validated a reduced-gravity-specific miRNA signature in both mice and rats and separated the miRNAs associated with reduced gravity from those that associated with exposure to simulated deep space radiation in mice. The only observed exception was miR-217, which did not change. A similar miRNA dose response was observed based on correlation with muscle mass: the miRNAs of interest increased with reduced muscle mass (which is associated with lower simulated gravity both in this rat model and in astronauts). In addition, we have data that shows the impact of these miRNAs at low Earth orbit (LEO) from serum in mice that were flown to the International Space Station (ISS) for 30 days. This data is available in the related publication and will not be shown here to save space. To understand the specificity and variation of spaceflight-relevant miRNA levels within the global miRNA population, miRNA-seq was performed on tissue samples from 16 week old female C57BL/6J mice after exposure to simulated spaceflight conditions: normal loading (NL) or hindlimb unloading (HU) for two weeks prior to irradiation with 5 Gy gamma, 1 Gy simulated solar particle event (simSPE), 0.5 Gy simulated simplified galactic cosmic rays (SimGCRSim), or sham control. SimGCRSim consist of 5 ions: protons at 1000MeV, 28Si at 600 MeV/n, 4He at 250 MeV/n, 16O at 350 MeV/n, 56Fe at 600 MeV/n, and protons at 250 MeV. This dose of radiation is equivalent to what an astronaut is predicted to receive in deep space during a Mars mission, though it is modeled as a single exposure over 25 minutes instead of the actual chronic exposures over 1.5 years. SimSPE consists of protons from 50 MeV to 150 MeV. 24 hours after irradiation, the mice were sacrificed and miRNA-seq was performed on tissue from plasma, heart, liver, and soleus muscle. From the miRNA-seq data we were able to uncover a common miRNA-seq profile. With overlapping existing miRNA family members with the original miRNA spaceflight signature. In addition, the miRNA spaceflight signature also has appeared in various tissues for these mice. Lastly, based on the global pathway analysis on the miRNA-seq data we observed global regulation of downstream targets that overlaps with the miRNA spaceflight signature. This work was done with Dr. Elizabeth Blaber (Rensselaer Polytechnic Institute grant 80NSSC20K0307). In collaboration with Dr. Christopher Mason, we also were able to confirm the presence of this spaceflight miRNA signature in humans from data available on the NASA Twins Study. In addition, in collaboration with Dr. Sylvain Costes, we also confirmed the presence of this miRNA signature with human peripheral blood mononuclear cells (PBMCs) that were irradiated with 0.3Gy 600MeV/n 56Fe particles. From the NASA Twins Study data we analyzed miRNA-sequence data, single-cell RNA sequencing (scRNA-seq), and single-cell sequencing Assay for Transposase-Accessible Chromatin (scATAC-seq) on the blood and different cell types from the blood. The miRNA downstream targets were determined in different cell types from the scRNA-seq and scATAC-seq data. Figure for this data is available in the publication in Cell Reports with the reference available in the Bibliography section. Finally, we have developed and utilized a mature human 3D microvascular tissue model seeded with human umbilical vein endothelial cells (HUVECs) to examine the molecular, cellular, and physiological effects of candidate miRNAs on cardiovascular damage. This work was done with Dr. Peter Grabham (Columbia University grant 80NSSC19K1658).

	Several of the candidate miRNAs showed marked increases at 48 hours after 0.5 Gy irradiation with simulated 5-ion simplified galactic cosmic rays (SimGCRSim) irradiation compared to controls, though the increases were not significant, possibly due to low sample numbers. Note that this dose of irradiation is equivalent to what an astronaut will receive in deep space Three of the increased miRNAs (miR-125, miR-16, and let-7a) have previously been predicted to be associated with space radiation induced cardiovascular damage, and were consequently chosen to be inhibited using antagomirs. In addition, we observed irradiation-mediated loss of vascularization as quantified by decreased green fluorescent protein (GFP) signal from GFP stained HUVECs, indicating that the vessels had shrunk or collapsed due to GCR irradiation. The impact of the space radiation was rescued by applying three antagomirs to inhibit miR-125, miR-16, and let-7a at 24 hours prior IR and 48 hours and 72 hours post IR.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Afshinnekoo E, Scott RT, MacKay MJ, Pariset E, Cekanaviciute E, Barker R, Gilroy S, Hassane D, Smith SM, Zwart SR, Nelman-Gonzalez M, Crucian BE, Ponomarev SA, Orlov OI, Shiba D, Muratani M, Yamamoto M, Richards SE, Vaishampayan PA, Meydan C, Foox J, Myrrhe J, Istasse E, Singh N, Venkateswaran K, Keune JA, Ray HE, Basner M, Miller J, Vitaterna MH, Taylor DM, Wallace D, Rubins K, Bailey SM, Grabham P, Costes SV, Mason CE, Beheshti A. "Fundamental biological features of spaceflight: Advancing the field to enable deep-space exploration." Cell. 2020 Nov 25;183(5):1162-84. Review. Erratum in: Cell. 2021 Nov 24;184(24):6002. https://doi.org/10.1016/j.cell.2020.10.050 ; PMID: 33242416; PMCID: PMC8441988 , Nov-2020
Articles in Peer-reviewed Journals	Malkani S, Chin CR, Cekanaviciute E, Mortreux M, Okinula H, Tarbier M, Schreurs AS, Shirazi-Fard Y, Tahimic CGT, Rodriguez DN, Sexton BS, Butler D, Verma A, Bezdan D, Durmaz C, MacKay M, Melnick A, Meydan C, Li S, Garrett-Bakelman F, Fromm B, Afshinnekoo E, Langhorst BW, Dimalanta ET, Cheng-Campbell M, Blaber E, Schisler JC, Vanderburg C, Friedländer MR, McDonald JT, Costes SV, Rutkove S, Grabham P, Mason CE, Beheshti A. "Circulating miRNA spaceflight signature reveals targets for countermeasure development." Cell Rep. 2020 Dec 8;33(10):108448. https://doi.org/10.1016/j.celrep.2020.108448 ; PMID: 33242410 , Dec-2020
Articles in Peer-reviewed Journals	Vanderburg C, Beheshti A. "MicroRNAs (miRNAs), the final frontier: The hidden master regulators impacting biological response in all organisms due to spaceflight. " THREE. 2020 Mar 9. https://three.jsc.nasa.gov/articles/miRNA_Beheshti.cfm , Mar-2020
Journal/Magazine covers	Afshinnekoo E, Scott RT, MacKay MJ, Pariset E, Cekanaviciute E, Barker R, Gilroy S, Hassane D, Smith SM, Zwart SR, Nelman-Gonzalez M, Crucian BE, Ponomarev SA, Orlov OI, Shiba D, Muratani M, Yamamoto M, Richards SE, Vaishampayan PA, Meydan C, Foox J, Myrrhe J, Istasse E, Singh N, Venkateswaran K, Keune JA, Ray HE, Basner M, Miller J, Vitaterna MH, Taylor DM, Wallace D, Rubins K, Bailey SM, Grabham P, Costes SV, Mason CE, Beheshti A. "Cell (cover): “The city lights spanning Japan, glimmering below Earth’s thin atmosphere, as well as the International Space Station with a docked Soyuz Space Capsule.” – Photograph by Scott Kelly. Article title is 'Fundamental biological features of spaceflight: Advancing the field to enable deep-space exploration.' " Cell. 2020 Nov 25;183(5).The article is featured on the cover of the journal at: https://www.cell.com/cell/issue?pii=S0092-8674(19)X0025-2#closeFullCover , Nov-2020
Journal/Magazine covers	Malkani S, Chin CR, Cekanaviciute E, Mortreux M, Okinula H, Tarbier M, Schreurs AS, Shirazi-Fard Y, Tahimic CGT, Rodriguez DN, Sexton BS, Butler D, Verma A, Bezdan D, Durmaz C, MacKay M, Melnick A, Meydan C, Li S, Garrett-Bakelman F, Fromm B, Afshinnekoo E, Langhorst BW, Dimalanta ET, Cheng-Campbell M, Blaber E, Schisler JC, Vanderburg C, Friedländer MR, McDonald JT, Costes SV, Rutkove S, Grabham P, Mason CE, Beheshti A. "Cell Reports (cover): “Day 142 during the Year-Long Mission of Scott Kelly, with the Aurora Borealis behind the docked Soyuz space capsule.” - Photograph by Scott Kelly. Article title is 'Circulating miRNA spaceflight signature reveals targets for countermeasure development.' " Cell Rep. 2020 Dec 8;33(10). https://www.cell.com/cell-reports/issue?pii=S2211-1247(19)X0050-9 , Dec-2020
Significant Media Coverage	Afshinnekoo E, Scott RT, MacKay MJ, Pariset E, Cekanaviciute E, Barker R, Gilroy S, Hassane D, Smith SM, Zwart SR, Nelman-Gonzalez M, Crucian BE, Ponomarev SA, Orlov OI, Shiba D, Muratani M, Yamamoto M, Richards SE, Vaishampayan PA, Meydan C, Foox J, Myrrhe J, Istasse E, Singh N, Venkateswaran K, Keune JA, Ray HE, Basner M, Miller J, Vitaterna MH, Taylor DM, Wallace D, Rubins K, Bailey SM, Grabham P, Costes SV, Mason CE, Beheshti A. "Review of Space Biology in Cell. Media coverage on this and the whole package of papers in Cell Press on The Biology of Spaceflight." CNN, Forbes, Popular Science, Engadget, Chemical & Engineering News, MIT Technology Review, November 2020., Nov-2020